

Dang Nguyen

Email: dangnguyen@uchicago.edu | Website: dangng2004.github.io/ | Phone: +1 (773) 936-0857

Education

2024 - Present	Computer Science Ph.D. , The University of Chicago. Advisor: Prof. Chenhao Tan
2020 - 2024	Computer Science B.S. & Mathematics B.A. , The University of Chicago.

Selected Publications

2025	On the Effectiveness and Generalization of Race Representations for Debiasing High-Stakes Decisions. [Link] Nguyen, D. , & Tan, C. Pre-print. Under review. {fairness and bias, mechanistic interpretability, causal interventions}
2025	GPT-4V Cannot Generate Radiology Reports Yet. [Link] Jiang, Y.*, Chen, C.*, Nguyen, D. , Mervak, B., & Tan, C. In Findings of the Association for Computational Linguistics (NAACL). {evaluation of large vision-language models, prompt engineering, error analysis}
2023	Pragmatic Radiology Report Generation. [Link] Nguyen, D. , Chen, C., He, H., & Tan, C. In Proceedings of Machine Learning for Health (ML4H). {vision-language modeling, finetuning LLMs, reducing hallucinations}

Professional Experience

2022	NetApp Inc. , Research Intern. - Developed machine learning algorithms to detect slow disk drives. - Optimized an algorithm based on KL divergence and improved NetApp’s threshold-based detector’s true positive rate from 10% to 59%. - Used Apache Hive and SQL to create 3 novel datasets of slow disks.
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Research Experience

2023 - 2024	Chicago Human+AI Lab , Undergraduate Researcher. Advisor: Prof. Chenhao Tan Project title: On the Effectiveness and Generalization of Race Representations for Debiasing High-Stakes Decisions. - Designed synthetic datasets to evaluate racial biases in Gemma 2B Instruct and LLaMA 3.1 8B Instruct’s decisions in college admissions and hiring. - Used Distributed Alignment Search to find models’ causal race representations with up to 90% accuracy. - Reduced models’ biases by up to 57% using representation-based interventions. - Showed that race representations can be prompt-specific, highlighting the challenge of limited generalizability. Project title: Pragmatic Radiology Report Generation. - Studied automated chest X-ray report generation using multi-modal LLMs. - Used a Chi-squared test to show that the indication section of a report drives mentions of negative findings. - Prompted Flan-T5 to clean 200,000 reports, reducing model hallucinations by 69.5% from the previous state-of-the-art (SoTA). - Developed a vision-language model using ResNet and LLaMA 7B that outperforms the previous SoTA by 29% on disease identification, 43% in writing style, and 52.9% on negative findings reporting.
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- 2020 - 2022 **Machine Learning for Systems Research (UCARE Lab)**, Undergraduate Researcher.
Advisor: Prof. Haryadi Gunawi
- The project aimed to develop a machine learning model for slow disk detection.
 - Built a pipeline to retrieve and parse disk performance data from a vendor's database, and optimized it by 90%.
 - Applied K-means and DBSCAN on 23 out of 91 pairwise correlations of data features to detect anomalous disk clusters.
 - Developed an algorithm using KL divergence to detect slow disks and analyzed 50,000 disks in 3 days using 3 high-performance compute servers.

Teaching Experience

- Winter 2025 **Teaching Assistant**, CMSC 25700: Natural Language Processing, University of Chicago.
- Designed homework assignments on transformers, pre-training, and interpretability.
 - Lead a tutorial session on PyTorch.
 - Mentored 7 student teams on their final project over the course of 5 weeks.
 - 3 of my teams ranked in the top 6 out of 26 teams, and 1 team won the best final report award.